

On the Existence of a Solution to the Coupled HJB–KFP System

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Abstract

This report studies a stationary mean field game system given by a Hamilton–Jacobi–Bellman equation coupled with a Kolmogorov–Fokker–Planck equation. Under suitable assumptions, we establish the existence of a weak solution pair in $H_0^1(\Omega) \times H_0^1(\Omega)$, and we extend the argument to a variant with a non-linear source term in the KFP equation.

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1 Introduction

Context and Motivation

Mean field games is a branch of game theory devoted to the analysis of a special subset of games. These systems are distinguished by having a very large number of players, each of whom is considered almost insignificant, with minimal influence on the system as a whole. In measure theory this corresponds to nullifying sets with countable measure and thinking of functions in the distributional sense.

Furthermore, each player is a rational observer and the general trend around them informs their decision on how to act, with the aim of achieving an optimal outcome. The famous example in economics of such a game, introduced by Aumann, is the behaviour of stock traders whom we treat as a continuum [1]. In this paper we consider the stationary equilibrium problem characterised by a coupled pair of equations: a Hamilton–Jacobi–Bellman (HJB) equation and a Kolmogorov–Fokker–Planck (KFP) equation.

$$\begin{aligned} -\Delta u + H(x, \nabla u) + \lambda u &= f(x, m) \quad \text{in } \Omega \text{ HJB,} \\ -\Delta m - \operatorname{div}\left(m \frac{\partial H}{\partial p}(x, \nabla u)\right) + \lambda m &= G(x) \quad \text{in } \Omega \text{ KFP.} \end{aligned} \tag{1.1}$$

This is supplemented with homogeneous Dirichlet boundary conditions $u = 0$ and $m = 0$ on $\partial\Omega$.

Interpretation of Equations

Here $m(x)$ represents the player density and $u(x)$ the value function of a representative agent in the mean field game. In other words, $u(x)$ is a player's optimal solution achieved by minimising the expected cost at position x . The term $\frac{\partial H}{\partial p}$ denotes the partial derivative of H with respect to the momentum variable (i.e. the ∇u argument), and $\lambda > 0$. HJB describes a representative player's optimal control behaviour for a given player density m .

On the other hand, KFP gives the evolution of $m(\cdot)$ when every player satisfies the optimal solution u . Each player tries to minimise their cost function $f(x, m)$ (in HJB), and the optimal control for an individual player is given by $\alpha^* = -\frac{\partial H}{\partial p}(x, \nabla u)$, which makes up the drift term in KFP. The Laplacian terms are diffusion terms accounting for randomness (e.g. Brownian motion) and preventing perfect alignment of players even under a shared strategy. Finally, $G(\cdot)$ is an exogenous term controlling entry/exit of players and therefore directly affects m in KFP. [4, 2]

Mathematical Contribution

In this paper we analyse the coupled system equation (1.1) and investigate the existence of a weak solution pair $(u, m) \in H_0^1(\Omega) \times H_0^1(\Omega)$. Establishing the existence of a solution pair (u, m) is important because such solutions correspond to Nash equilibria of the mean field game. The function u determines the optimal strategy of a representative agent, while m

describes the distribution of agents when all players adopt this strategy. A solution of the coupled system therefore represents a self-consistent equilibrium of the game. In addition, we consider a more general formulation of the system and prove an existence result in this extended setting, which constitutes an original contribution of this work.

Assumptions

1. Ω is bounded and $\partial\Omega$ is C^1 .
2. $H(x, \nabla u) : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ is
 - Lipschitz continuous with respect to ∇u with Lipschitz constant $\tilde{C}$,
 - bounded by $C(|\nabla u| + 1)$ for a constant C , for all $x \in \mathbb{R}$ and $\nabla u \in \mathbb{R}^n$.
3. The partial derivative $\frac{\partial H}{\partial p}(x, p)$ is continuous and essentially bounded by some constant L .
4. λ is sufficiently large (in particular $\lambda > \max(\frac{L^2}{4}, \frac{C^2}{2}, \frac{\tilde{C}^2}{4})$).
5. $f(x, m(x)) : \Omega \times L^2(\Omega) \rightarrow L^2(\Omega)$ and is continuous with respect to m .
6. $G(x) : \Omega \rightarrow L^2(\Omega)$.

Example 1.1. A Hamiltonian satisfying Assumption 2 is

$$H(x, p) = \sqrt{1 + |p|^2} + V(x) \quad \text{on } \Omega = B(0, 1) \text{ in } \mathbb{R}^n,$$

where $V \in L^\infty(\Omega)$. In this case, we get the associated drift

$$\frac{\partial H}{\partial p} = \frac{p}{\sqrt{1 + |p|^2}}.$$

Example 1.2. For a coupling term satisfying Assumption 5, one may consider

$$f(x, m) = a(x) \phi(m),$$

where $a(x) \in L^\infty(\Omega)$ and $\phi : L^2(\Omega) \rightarrow L^2(\Omega)$ is continuous.

Main Result

Theorem 1.3 (Existence of the MFG solution). *Let the Mean Field Game system be characterised by equation (1.1) and suppose the assumptions above hold. Then a solution $(u, m) \in H_0^1(\Omega) \times H_0^1(\Omega)$ exists.*

We prove this theorem throughout the rest of this paper.

2 Method

Our proof consists of two stages. First, we demonstrate solvability for each equation separately, while fixing the other variable (HJB with fixed m , and KFP with fixed u). Second, we show the existence of a solution pair solving both equations simultaneously. Before starting our proof it is important to introduce the following inequalities, which will be used repeatedly and will lend themselves to many proofs throughout this paper.

Definition 2.1. Parameters $p, q \in [1, \infty]$ are *conjugate* if $\frac{1}{p} + \frac{1}{q} = 1$.

Theorem 2.2 (Hölder's inequality). *Let $f \in L^p(\Omega)$ and $g \in L^q(\Omega)$ where p, q are conjugate. Then*

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

Remark 2.3. In particular, when $p = q = 2$ we obtain the Cauchy–Schwarz inequality

$$\|fg\|_1 \leq \|f\|_2 \|g\|_2.$$

Theorem 2.4 (Young's inequality). *Let $a, b \in \mathbb{R}$. Then*

$$2ab \leq \frac{a^2}{\varepsilon^2} + \varepsilon^2 b^2 \quad \forall \varepsilon > 0.$$

2.1 Independent unique solutions

2.1.1 Solving Kolmogorov–Fokker–Planck

To solve

$$-\Delta m - \operatorname{div}\left(m \frac{\partial H}{\partial p}(x, \nabla u)\right) + \lambda m = G(x)$$

with fixed u , we note that the equation is linear in $m(x)$, so we appeal to the Lax–Milgram theorem [5].

Theorem 2.5 (Lax–Milgram). *Let V be a Hilbert space and let $a : V \times V \rightarrow \mathbb{R}$ be a bilinear form such that for all $u, v \in V$,*

$$(i) \text{ (boundedness) } |a(u, v)| \leq c_1 \|u\| \|v\|,$$

$$(ii) \text{ (coercivity) } a(v, v) \geq c_0 \|v\|^2.$$

Let $l : V \rightarrow \mathbb{R}$ be a bounded linear functional. Then there exists a unique $u \in V$ such that

$$a(u, v) = l(v) \quad \text{for all } v \in V.$$

Moreover, $\|u\|_V \leq \frac{1}{c_0} \|l\|_{V^}$.*

$$a(m, v) = \int_{\Omega} \nabla m \cdot \nabla v + m \nabla v \cdot \frac{\partial H}{\partial p}(x, \nabla u) + \lambda m v \, dx, \quad l(v) = \int_{\Omega} G(x) v \, dx. \quad (2.1)$$

We now show that KFP meets the conditions of Theorem 2.5.

Proposition 2.6 (Boundedness of KFP). *The form $a(m, v)$ is bounded on $H_0^1(\Omega) \times H_0^1(\Omega)$.*

Proof. For arbitrary $m, v \in H_0^1(\Omega)$,

$$\begin{aligned} |a(m, v)| &\leq \int_{\Omega} |\nabla m| |\nabla v| + |\nabla v| |m| \left| \frac{\partial H}{\partial p} \right| + \lambda |m| |v| \, dx \\ &\leq \|\nabla m\|_{L^2} \|\nabla v\|_{L^2} + L \|m\|_{L^2} \|\nabla v\|_{L^2} + \lambda \|m\|_{L^2} \|v\|_{L^2}. \end{aligned}$$

By Cauchy–Schwarz and Assumption 3, there exists $C > 0$ such that

$$|a(m, v)| \leq C \|m\|_{H^1} \|v\|_{H^1}.$$

Thus $a(m, v)$ is bounded. $\square$

Proposition 2.7 (Coercivity of KFP). *If $\lambda > 0$ is sufficiently large, then $a(\cdot, \cdot)$ is coercive.*

Proof. Consider

$$\begin{aligned} a(v, v) &= \int_{\Omega} |\nabla v|^2 + v \nabla v \cdot \frac{\partial H}{\partial p}(x, \nabla u) + \lambda v^2 \, dx \\ &\geq \|\nabla v\|_{L^2}^2 - L \|v\|_{L^2} \|\nabla v\|_{L^2} + \lambda \|v\|_{L^2}^2. \end{aligned}$$

Applying Young’s inequality yields

$$a(v, v) \geq \left(1 - \frac{\varepsilon^2}{2}\right) \|\nabla v\|_{L^2}^2 + \left(\lambda - \frac{L^2}{2\varepsilon^2}\right) \|v\|_{L^2}^2, \quad \forall \varepsilon \in \mathbb{R}.$$

To ensure $1 - \varepsilon^2/2 > 0$, impose $\varepsilon^2 < 2$. Choosing $\varepsilon^2 = 2$ (as a limiting choice) gives the weakest requirement

$$\lambda > \frac{L^2}{4},$$

which ensures coercivity. $\square$

Theorem 2.8 (Existence of unique KFP solution). *Let the Kolmogorov–Fokker–Planck equation be as in equation (1.1) and suppose that $\lambda > 0$ is sufficiently large. Then for any fixed $u \in H_0^1(\Omega)$, there exists a unique $m \in H_0^1(\Omega)$ solving the equation.*

Proof. Let $a(m, v) = l(v)$ be the weak formulation as in equation (2.1). By Propositions 2.6 and 2.7, $a(\cdot, \cdot)$ is bounded and coercive. It remains to show l is bounded. By Cauchy–Schwarz,

$$\|l\| = \sup_{\|v\|_{H^1} \leq 1} \left| \int_{\Omega} G v \, dx \right| \leq \sup_{\|v\|_{L^2} \leq 1} \|v\|_{L^2} \|G\|_{L^2} \leq \|G\|_{L^2(\Omega)}.$$

Thus l is bounded. By Theorem 2.5, there exists a unique $m \in H_0^1(\Omega)$. $\square$

Remark 2.9. Lax–Milgram applies here because KFP is linear in m once $u \in H_0^1(\Omega)$ is fixed.

2.1.2 Solving Hamilton–Jacobi–Bellman

Lax–Milgram does not apply directly to

$$-\Delta u + H(x, \nabla u) + \lambda u = f(x, m)$$

with fixed m , because the equation is non-linear so we resort to a fixed point method instead. The idea, is to create some kind of map from HJB with the right properties such that it has a fixed point and for that very fixed point to be our solution. Remarkably, this fixed point method is general enough that we can use it to solve for existence in the system as whole. Indeed, this will be the concern of the next subsection. Throughout this section, X denotes a real Banach space. We now introduce the fixed point theorems that rely on suitable properties for the HJB context.

Theorem 2.10 (Schauder’s fixed point theorem). *If $K \subset X$ is compact and convex, and $A : K \rightarrow K$ is continuous, then A has a fixed point in K .*

Definition 2.11. A mapping $A : X \rightarrow X$ is called *compact* if for every bounded sequence $(u_k)_{k=1}^\infty$, the sequence $(A[u_k])_{k=1}^\infty$ is precompact in X .

Theorem 2.12 (Schaefer’s fixed point theorem). *Suppose $A : X \rightarrow X$ is continuous and compact, and assume the set*

$$\{u \in X : u = \mu A[u] \text{ for some } 0 \leq \mu \leq 1\}$$

*is bounded. Then A has a fixed point.*¹

Let us now design the map whose fixed point is a solution to HJB. Consider

$$A : H_0^1(\Omega) \rightarrow H_0^1(\Omega), \quad A[u] = w \quad \text{where} \quad -\Delta w + \lambda w = f(x, m) - H(x, \nabla u).$$

We know w exists by Lax–Milgram Theorem 2.5, so A is well defined.² Furthermore by regularity theory³ we have

$$\begin{aligned} \|w\|_{H^2(\Omega)} &\leq C \|f(x, m) - H(x, \nabla u)\|_{L^2(\Omega)} \\ &\leq C \|f(x, m)\|_{L^2(\Omega)} + C \|H(x, \nabla u)\|_{L^2(\Omega)} \\ &\leq C \|f(x, m)\|_{L^2(\Omega)} + \hat{C} \|\nabla u\| + 1 \|1\|_{L^2(\Omega)} \quad (\text{Assumption 2}) \end{aligned}$$

Where $C, \hat{C}$ are constants - note that these constants are not the same as the ones appearing in our assumptions above. We now prove the existence of an HJB solution by way of our fixed point map.

¹See [3, Section 9.2] for proofs.

²LHS is bounded and coercive and RHS is bounded.

³See [3, Section 6.3]

Theorem 2.13 (Existence of HJB Solution). *If $\lambda > 0$ is sufficiently large, then for any fixed $m \in H_0^1(\Omega)$ there exists $u \in H^2(\Omega) \cap H_0^1(\Omega)$ solving HJB.*

Proof. ⁴ In other words, for sufficiently large $\lambda > 0$, A has a fixed point. We show first that A meets the right criteria for Schaefer's Theorem. We assert that A is continuous and compact.

If $u_k \rightarrow u$ in $H_0^1(\Omega)$ and we denote $w_k = A[u_k]$ then higher regularity tells us

$$\sup_k \|w_k\|_{H^2(\Omega)} < \infty$$

(i.e. (w_k) is a bounded sequence in $H^2(\Omega) \cap H_0^1(\Omega)$) and there exists a convergent subsequence $(w_{k_j})_{j=1}^\infty \rightarrow w$ in $H_0^1(\Omega)$.

Now the weak form of HJB with w_{k_j} is

$$\int_{\Omega} \nabla w_{k_j} \cdot \nabla v + \lambda w_{k_j} \cdot v \, dx = \int_{\Omega} (f(x, m) - H(x, \nabla u_{k_j})) \cdot v \, dx, \text{ for any } v \in H_0^1(\Omega)$$

and consequently

$$\int_{\Omega} \nabla w \cdot \nabla v + \lambda w \cdot v \, dx = \int_{\Omega} (f(x, m) - H(x, \nabla u)) \cdot v \, dx, \text{ for any } v \in H_0^1(\Omega)$$

by Lipschitz continuity of H (Assumption 2). Thus, $w = A[u]$. Hence, $A[u_k] \rightarrow A[u]$ in $H_0^1(\Omega)$, so A is continuous. A is compact by similar reasoning since a bounded sequence $(u_k)_{k=1}^\infty$ yields a bounded sequence $(A[u_k])_{k=1}^\infty$ in $H^2(\Omega)$ which has a strongly convergent subsequence in $H_0^1(\Omega)$.

It remains to show

$$\{u \in H_0^1(\Omega) : u = \mu A[u] \text{ for some } 0 \leq \mu \leq 1\}$$

is bounded.

Substituting $w = u/\mu$ and setting $v = u$, we obtain

$$\begin{aligned} \int_{\Omega} |\nabla u|^2 + \lambda |u|^2 \, dx &= \mu \int_{\Omega} f u - H(x, \nabla u) u \, dx \\ &\leq \int_{\Omega} |f u| + |H(x, \nabla u) u| \, dx, \quad \text{since } 0 \leq \mu \leq 1 \\ &\leq \int_{\Omega} |f u| + C(|\nabla u| + 1)|u| \, dx \quad (\text{Assumption 2}) \end{aligned}$$

To isolate $|\nabla u|^2$ and u^2 terms on the RHS, we apply Young's inequality (Theorem 2.4) once

⁴Proof is adapted from [3, Section 9.2]

on $|fu|$ and twice over on $C(|\nabla u| + 1)|u|$, to obtain

$$|fu| \leq \frac{1}{2} \left(\frac{f^2}{\varepsilon^2} + \varepsilon^2 u^2 \right) \quad \varepsilon \in (0, \infty)$$

and

$$\begin{aligned} C(|\nabla u| + 1)|u| &\leq \frac{1}{2} \left(\frac{1}{\delta^2} |Cu|^2 + \delta^2 (|\nabla u| + 1)^2 \right), \quad \delta \in (0, 1) \\ &\leq \frac{1}{2} \left(\frac{1}{\delta^2} |Cu|^2 + \delta^2 (1 + \gamma^2) |\nabla u|^2 + \delta^2 (1 + \frac{1}{\gamma^2}) \right) \\ &= \frac{1}{2} \left(\frac{1}{\delta^2} |Cu|^2 + 2\delta^2 |\nabla u|^2 + 2\delta^2 \right) \quad \text{WLOG setting } \gamma = 1. \end{aligned}$$

Returning to the weak form inequality we get

$$\int_{\Omega} |\nabla u|^2 + \lambda |u|^2 dx \leq \frac{1}{2} \int_{\Omega} \frac{f^2}{\varepsilon^2} + \left(\varepsilon^2 + \frac{C^2}{\delta^2} \right) u^2 + 2\delta^2 |\nabla u|^2 + 2\delta^2 dx.$$

Provided $\lambda > \frac{C^2}{2}$ there exist sufficiently small $\varepsilon > 0$ and sufficiently large $\delta < 1$ such that

$$\lambda > \frac{1}{2} \left(\varepsilon^2 + \frac{C^2}{\delta^2} \right) \text{ and } 1 > \delta^2$$

hold simultaneously. Taking the difference of both sides and denoting the constant factor as $K := \int_{\Omega} \delta^2 dx$, yields

$$\alpha \int_{\Omega} |\nabla u|^2 + |u|^2 dx \leq \frac{1}{2\varepsilon^2} \int_{\Omega} f(x, m)^2 dx + K,$$

where α is the minimum of both factors of u^2 and ∇u^2 . Equivalently,

$$\alpha \|u\|_{H_0^1(\Omega)} \leq \frac{1}{2\varepsilon^2} \|f(x, m)\|_{L^2(\Omega)} + K.$$

Thus if $\lambda > \frac{C^2}{2}$, the set

$$\{u \in H_0^1(\Omega) : u = \mu A[u] \text{ for some } 0 \leq \mu \leq 1\}$$

is bounded independently of μ . By Theorem 2.12, A has a fixed point and thus HJB has a solution. $\square$

Remark 2.14. We can always tune ε, δ and γ for given $\lambda > \frac{C^2}{2}$ to satisfy boundedness. This utilises the strictness of the inequality in the condition on λ .

Proposition 2.15. *If $\lambda > 0$ is sufficiently large, then the fixed point u is unique.*

Proof. Suppose u_1, u_2 both solve HJB for a given $m \in H_0^1(\Omega)$. Taking the weak difference

and setting $v = u_1 - u_2$ gives

$$\begin{aligned} \|\nabla(u_1 - u_2)\|_{L^2}^2 + \lambda \|u_1 - u_2\|_{L^2}^2 &\leq \int_{\Omega} |u_1 - u_2| |H_1 - H_2| dx \\ &\leq \tilde{C} \|u_1 - u_2\|_{L^2} \|\nabla(u_1 - u_2)\|_{L^2} \end{aligned}$$

using Assumption 2 and Cauchy–Schwarz. Hence, by Young’s inequality,

$$\|\nabla(u_1 - u_2)\|_{L^2}^2 + \lambda \|u_1 - u_2\|_{L^2}^2 \leq \frac{1}{2} \left(\frac{\tilde{C}^2}{\varepsilon^2} \|u_1 - u_2\|_{L^2}^2 + \varepsilon^2 \|\nabla(u_1 - u_2)\|_{L^2}^2 \right) \quad \varepsilon \in (0, \sqrt{2}).$$

Thus by the same reasoning as when we proved existence,

$$\lambda > \frac{\tilde{C}^2}{4} \implies u_1 = u_2 \text{ in } L^2(\Omega) \iff u_1 = u_2 \text{ in } H_0^1(\Omega).$$

□

The following theorem summarises our results.

Theorem 2.16 (Existence of unique HJB solution). *Let the Hamilton-Jacobi equation be as in equation (1.1) and suppose that $\lambda > 0$ is sufficiently large. Then for any fixed $m \in H_0^1(\Omega)$, there exists a unique $u \in H_0^1(\Omega)$ solving the equation.*

Proof. There exists a solution by Theorem 2.13 and it is unique by Proposition 2.15. □

2.2 Solving the Coupled System of Equations

Having solved the two equations in equation (1.1) in isolation, we now look to solve them as a system. As stated above, the fixed point theorems will allow us to prove existence of a solution pair for the system as a whole. The challenge, is to define and prove the existence of a fixed point for a suitable map. Our work in the previous subsection allows us to define the following maps.

Definition 2.17. Define $S : H_0^1(\Omega) \rightarrow H_0^1(\Omega)$ by $S(u) = m$, where m solves KFP for given u .

Definition 2.18. Define $T : H_0^1(\Omega) \rightarrow H_0^1(\Omega)$ by $T(m) = u$, where u solves HJB for given m .

Definition 2.19. Let T and S be as in definitions definition 2.17 and definition 2.18. Define

$$A : H_0^1(\Omega) \rightarrow H_0^1(\Omega), \quad A := S \circ T.$$

That is, A takes $m \in H_0^1(\Omega)$, converts it to $u = T(m)$ solving HJB, and then produces $\tilde{m} = S(u)$ solving KFP. If A has a fixed point, then $m = \tilde{m}$, and the coupled system has a solution pair.

We verify the hypotheses of Schaefer's theorem.

Lemma 2.20. *Let A be defined as in definition 2.19 and suppose that $\lambda > 0$ as it appears in equation (1.1) is sufficiently large. Then, A is continuous.*

Since A is simply a composition of S and T , it suffices to show they both continuous.

Proposition 2.21. *If $\lambda > 0$ is sufficiently large, T is continuous.*

Proof. Let $(m_k)_{k=1}^\infty \rightarrow m$ in $H_0^1(\Omega)$. Denote $u = T[m]$, $u_k = T[m_k]$, $H = H(x, \nabla u)$, $H_k = H(x, \nabla u_k)$ and $f_k = f(x, m_k)$. Taking the weak formulation of the difference in HJB gives

$$\begin{aligned} \|\nabla(u - u_k)\|_{L^2}^2 + \lambda \|u - u_k\|_{L^2}^2 &\leq \int_{\Omega} |u - u_k| |H - H_k| + |u - u_k| |f - f_k| dx \\ &\leq \tilde{C} \|u - u_k\|_{L^2} \|\nabla(u - u_k)\|_{L^2} + \|u - u_k\|_{L^2} \|f - f_k\|_{L^2}. \end{aligned} \quad (1)$$

By Cauchy-Schwarz and Assumption 2. Since f is $L^2(\Omega)$ -continuous (Assumption 5) and

$$m_k \rightarrow m \text{ in } H_0^1(\Omega) \implies m_k \rightarrow m \text{ in } L^2(\Omega),$$

$\|f - f_k\|_{L^2} \rightarrow 0$. Also, from the energy estimate

$$\|\nabla u_k\|_{L^2}^2 + \lambda \|u_k\|_{L^2}^2 \leq \int_{\Omega} |u_k| |f_k| + |u_k| |H_k| dx.$$

Adopting the same argument for boundedness as in Theorem 2.13 gives the inequality

$$\alpha \|u_k\|_{H_0^1(\Omega)} \leq \frac{1}{2\varepsilon^2} \|f_k\|_{L^2(\Omega)} + K.$$

where K , m and ε are independent constants. By the fact that $f_k \rightarrow f$ we know that $\|f_k\|_{L^2(\Omega)}$ is bounded and so therefore (u_k) is bounded in $H_0^1(\Omega)$. Thus, $\|u - u_k\|_{L^2} \|f - f_k\|_{L^2} \rightarrow 0$. Therefore, in the limit of k , (1) becomes

$$\|\nabla(u - u_k)\|_{L^2}^2 + \lambda \|u - u_k\|_{L^2}^2 \leq \tilde{C} \|u - u_k\|_{L^2} \|\nabla(u - u_k)\|_{L^2}$$

for which $\lambda > 0$ being sufficiently large implies $u_k \rightarrow u$ i.e T is continuous. ⁵ □

Proposition 2.22. *If $\lambda > 0$ is sufficiently large, S is continuous.*

Proof. Let $(u_k)_{k=1}^\infty \rightarrow u$ in $H_0^1(\Omega)$. Denote $m = S[u]$, $m_k = S[u_k]$, $b = \frac{\partial H}{\partial p}(x, \nabla u)$ and

⁵See the uniqueness argument above (Proposition 2.15).

$b_k = \frac{\partial H}{\partial p}(x, \nabla u_k)$. Taking the weak formulation of the difference in KFP yields

$$\|\nabla(m - m_k)\|_{L^2}^2 + \lambda \|m - m_k\|_{L^2}^2 \leq \int_{\Omega} |\nabla(m - m_k)| |m b - m_k b_k| dx.$$

Define the correction vector field $\varepsilon_k := b - b_k$. Note that since b_k is continuous w.r.t. u_k we have $\lim_{k \rightarrow \infty} \|\varepsilon_k\|_{L^2(\Omega)} = 0$. Then,

$$\begin{aligned} \|\nabla(m - m_k)\|_{L^2}^2 + \lambda \|m - m_k\|_{L^2}^2 &\leq \int_{\Omega} |\nabla(m - m_k)| |m b - m_k b| + |\nabla(m - m_k)| |m_k| |\varepsilon_k| dx \\ &\leq L \int_{\Omega} |\nabla(m - m_k)| |m - m_k| dx + \int_{\Omega} |\nabla(m - m_k)| |m_k| |\varepsilon_k| dx. \end{aligned}$$

m and m_k are bounded in $H_0^1(\Omega)$ so as k gets large $|\nabla(m - m_k)| |m_k| |\varepsilon_k| \rightarrow 0$. Therefore, by the largeness of λ (see Proposition 2.7), we get $m_k \rightarrow m$ i.e S is continuous. $\square$

Remark 2.23. The image of S is always bounded as a direct consequence of Lax-Milgram (Theorem 2.5).

2.2.1 Compactness of A

Lemma 2.24. *Let A be defined as in definition 2.19 and suppose that $\lambda > 0$ as it appears in equation (1.1) is sufficiently large. Then, A is compact.*

Since S is continuous, it suffices to show that T is compact in $H_0^1(\Omega)$:

$$T \text{ compact} \implies A = S \circ T \text{ compact}.$$

We introduce the following theorems. These will allow us to conclude convergence of a subsequence in a bigger space if we are given a bounded sequence in a smaller space. The particular relationship that concerns us is the one between $L^2(\Omega)$ and $H_0^1(\Omega)$.

Definition 2.25. Let X and Y be Banach spaces with $X \subset Y$. We say that X is compactly embedded in Y , written $X \subset\subset Y$, provided:

- (i) $\|x\|_Y \leq C \|x\|_X$ for some constant C ,
- (ii) each bounded sequence in X is precompact in Y .

Theorem 2.26 (Rellich–Kondrachov compactness theorem). *Assume U is a bounded open subset of $\mathbb{R}^n$ and ∂U is C^1 . Suppose $1 < p < n$. Then*

$$W^{1,p}(U) \subset\subset L^q(U) \quad \text{for each } 1 \leq q < p^*, \quad p^* = \frac{np}{n-p}.$$

Remark 2.27. Observe $p^* > p$. In particular,

$$W^{1,p}(U) \subset\subset L^p(U) \quad \text{for all } 1 \leq p < n.$$

Example 2.28. The space we're interested in, is $H_0^1(\Omega)$. We know that $H_0^1(\Omega) \subset H^1(\Omega)$. Thus, by Rellich Kondrachov and Assumption 1

$$H_0^1(\Omega) \subset\subset L^2(\Omega).$$

Proposition 2.29. *If $\lambda > 0$ is sufficiently large, T is compact.*

Proof. Let $(m_k)_{k=1}^\infty$ be a bounded sequence in $H_0^1(\Omega)$. Since Ω satisfies Assumption 1, there exists a subsequence $(m_{k_j})_{j=1}^\infty$ converging in $L^2(\Omega)$ to some $m \in L^2(\Omega)$. We want to show $T(m_{k_j}) \rightarrow T(m)$. Remember the weak form of $T(m) - T(m_{k_j})$

$$\begin{aligned} \left\| \nabla(u - u_{k_j}) \right\|_{L^2}^2 + \lambda \left\| u - u_{k_j} \right\|_{L^2}^2 &\leq \int_{\Omega} |u - u_{k_j}| |H - H_{k_j}| + |u - u_{k_j}| |f - f_{k_j}| dx \\ &\leq \tilde{C} \left\| u - u_{k_j} \right\|_{L^2} \left\| \nabla(u - u_{k_j}) \right\|_{L^2} + \left\| u - u_{k_j} \right\|_{L^2} \left\| f - f_{k_j} \right\|_{L^2} \end{aligned}$$

Because f is $L^2(\Omega)$ -continuous (Assumption 5), the continuity argument for T applies with m_{k_j} . Thus T is compact, and hence A is compact. $\square$

We have shown that A is continuous and compact. The last step before we can apply Schauder's theorem is to prove that if there were to exist a fixed point, it would be bounded.

2.2.2 Boundedness of the Fixed Point Set

Consider

$$\{m \in H_0^1(\Omega) : m = \mu A[m] \text{ for some } \mu \in [0, 1]\}.$$

The corresponding system, with $A[m]$ substituted for m/μ , is

$$-\Delta u + H(x, \nabla u) + \lambda u = f(x, m), \quad \frac{1}{\mu} \left(-\Delta m - \operatorname{div} \left(m \frac{\partial H}{\partial p}(x, \nabla u) \right) + \lambda m \right) = G(x).$$

Rearranging the latter and using the Lax–Milgram bound implies

$$\|m\|_{L^2} \leq \frac{1}{c_0} \mu \|G\|_{L^2} \leq \frac{1}{c_0} \|G\|_{L^2},$$

which is bounded in $L^2(\Omega)$ (Assumption 6). By Schaefer's theorem, A has a fixed point. We may conclude our finding with the next theorem.

Proof of the Main Result

Proof of Theorem 1.3. Let A be as in definition 2.19. A is compact and continuous and has bounded fixed points (if they were to exist) by the above, Lemma 2.20 and Lemma 2.24. By Schaefer's theorem (Theorem 2.12), A has a fixed point. Thus, a solution pair $(u, m) \in H_0^1(\Omega) \times H_0^1(\Omega)$ to the MFG system exists. $\square$

2.3 Swapping Composition Order of T and S

In hindsight, we can see that the composition of T (definition 2.18) followed by S (definition 2.17) was convenient because it provided a natural bound for the fixed point. We relied

on the fact that the image of S is always bounded (Theorem 2.5) independent of m . An equally valid construction for A however, is the reverse composition

$$A := T \circ S.$$

Here too the existence of a fixed point proves Theorem 1.3 yet crucially, recall that we have not shown the image of T to have any natural bound in $H_0^1(\Omega)$. We note that bounded sets under S remain bounded and A retains its continuity and compactness.⁶ Therefore, we need only demonstrate boundedness of the fixed point set for us to use Schaefer's Theorem. Under this new construction consider

$$\{u \in H_0^1(\Omega) : u = \mu A[u] \text{ for some } \mu \in [0, 1]\}.$$

The corresponding system, with $A[u]$ substituted for u/μ , is

$$-\frac{\Delta u}{\mu} + H(x, \frac{\nabla u}{\mu}) + \frac{\lambda u}{\mu} = f(x, m), \quad -\Delta m - \operatorname{div}(m \frac{\partial H}{\partial p}(x, \nabla u)) + \lambda m = G(x).$$

We obtain the weak form of HJB with $v = u$

$$\begin{aligned} \int_{\Omega} |\nabla u|^2 + \lambda |u|^2 dx &= \mu \int_{\Omega} f u - H(x, \frac{\nabla u}{\mu}) u dx \\ &\leq \int_{\Omega} |f u| + \mu |H(x, \frac{\nabla u}{\mu}) u| dx \quad \text{since } 0 \leq \mu \leq 1 \\ &\leq \int_{\Omega} |f u| + \mu C(|\frac{\nabla u}{\mu}| + 1) |u| dx \quad (\text{Assumption 2}) \\ &= \int_{\Omega} |f u| + C(|\nabla u| + 1) |u| dx. \end{aligned}$$

This is the same form as the inequality in the proof of Theorem 2.13. There we arrived at the following bound

$$\alpha \|u\|_{H_0^1(\Omega)} \leq \frac{1}{2\varepsilon^2} \|f(x, m)\|_{L^2(\Omega)} + K$$

with α , ε and K independent constants. The bound of u must be independent of m so we introduce a growth condition on f .

Proposition 2.30. *Let $|f(x, m)| \leq (|m|+1)$ then $\{u \in H_0^1(\Omega) : u = \mu A[u] \text{ for some } \mu \in [0, 1]\}$ is bounded.*

Proof. We have

$$\begin{aligned} \alpha \|u\|_{H_0^1(\Omega)} &\leq \frac{1}{2\varepsilon^2} \|f(x, m)\|_{L^2(\Omega)} + K \\ &\leq \frac{1}{2\varepsilon^2} (\|m\|_{L^2(\Omega)} + |\Omega|) + K \end{aligned}$$

⁶Maps S and T themselves are unchanged. We have already shown that S is continuous (Proposition 2.22) and that T is continuous and compact (Propositions 2.21 and 2.29).

$$\begin{aligned}
&\leq \frac{1}{2\varepsilon^2} (\|m\|_{H_0^1(\Omega)} + |\Omega|) + K \\
&\leq \frac{1}{2\varepsilon^2} \left(\frac{1}{c_0} \|G\|_{L^2(\Omega)} + |\Omega| \right) + K
\end{aligned}$$

by the Lax-Milgram bound (Theorem 2.5). Thus, the set is bounded. $\square$

We summarise the analysis of this subsection with a lemma.

Lemma 2.31. *Let $A = T \circ S$ and suppose that $\lambda > 0$ as it appears in equation (1.1) is sufficiently large. Then,*

$$|f(x, m)| \leq (|m| + 1) \implies A \text{ has a fixed point.}$$

Proof. A is compact and continuous since T is compact and continuous (Proposition 2.29, Proposition 2.21) and S is bounded continuous (Proposition 2.22). The set of fixed points is bounded by Proposition 2.30. Thus, there exists a fixed point by Schaefer's Theorem. $\square$

3 Extension to Non-linear KFP

In certain applications it is natural to allow for the entry or exit of agents from the system. This can be represented by introducing a source term $G(x, m)$ in place of $G(x)$ in equation (1.1). Such terms may model, for example, external recruitment of agents into the population, departures from the system, or density-dependent entry and exit effects. In this section we consider a class of nonlinear functions $G(x, m)$ satisfying the structural conditions stated in Assumption 7. Our new system of equations reads

$$\begin{aligned}
-\Delta u + H(x, \nabla u) + \lambda u &= f(x, m) \quad \text{in } \Omega, \\
-\Delta m - \operatorname{div}\left(m \frac{\partial H}{\partial p}(x, \nabla u)\right) + \lambda m &= G(x, m) \quad \text{in } \Omega.
\end{aligned} \tag{3.1}$$

In this scenario we cannot apply the Lax-Milgram Theorem to KFP. We will therefore adopt fixed point theorems to find a solution but as with $H(x, \nabla u)$ we impose assumptions on $G(x, m)$.

7. $G(x, m) : \Omega \times L^2(\Omega) \rightarrow L^2(\Omega)$ is

- Lipschitz continuous with respect to m with Lipschitz constant $\tilde{B}$,
- bounded by $B(|m| + 1)$ for a given constant B , for all $x \in \Omega$ and $m \in L^2(\Omega)$.

8. In *addition* to our previous constraint on λ , we have $\lambda > \max(B, \tilde{B}) + \frac{L^2}{4}$

Example 3.1. For a source term G satisfying Assumption 7, one may consider

$$G(x, m) = a(x) \tanh(m),$$

where $a \in L^\infty(\Omega)$. From a modelling perspective, this choice corresponds to density-dependent entry or exit with saturation: for small m the response is approximately linear,

while for large m the source term remains bounded.

Theorem 3.2 (Existence of Non-linear KFP Solution). *If $\lambda > 0$ is sufficiently large, there exists $m \in H^2(\Omega) \cap H_0^1(\Omega)$ solving KFP with fixed u .*

Proof. We design the map whose fixed point is a solution to the non-linear equation.

$$A : H_0^1(\Omega) \rightarrow H_0^1(\Omega), \quad A[m] = w \quad \text{where} \quad -\Delta w - \operatorname{div}\left(w \frac{\partial H}{\partial p}(x, \nabla u)\right) + \lambda w = G(x, m).$$

We know w exists by Lax–Milgram Theorem 2.5, so A is well defined. Continuity and compactness are the same as in Theorem 2.13 so we omit them. The only difference is in the threshold of λ which we'll now derive. As in Theorem 2.13 we want to show

$$\{m \in H_0^1(\Omega) : m = \mu A[m] \text{ for some } 0 \leq \mu \leq 1\}$$

is bounded. Substituting $w = m/\mu$ and taking $v = m$, we obtain the weak form

$$\int_{\Omega} |\nabla m|^2 + m \nabla m \frac{\partial H}{\partial p} + \lambda |m|^2 dx = \mu \int_{\Omega} G(x, m) m dx$$

Moving the $\frac{\partial H}{\partial p}$ term over and Assumption 3 give

$$\begin{aligned} \int_{\Omega} |\nabla m|^2 + \lambda |m|^2 dx &\leq \int_{\Omega} |G(x, m)| |m| + L |m| |\nabla m| dx \\ &\leq \int_{\Omega} B(|m| + 1) |m| + L |m| |\nabla m| dx \quad (\text{Assumption 7}) \end{aligned}$$

To isolate $|\nabla m|^2$ and m^2 terms on the RHS, we apply Young's inequality 2.4 to give

$$\int_{\Omega} |\nabla m|^2 + \lambda |m|^2 dx \leq \int_{\Omega} \frac{1}{2} \delta^2 |\nabla m|^2 + \left(B + \frac{1}{2} \varepsilon^2 + \frac{L^2}{2\delta^2}\right) |m|^2 dx + K \quad \varepsilon, \delta \in (0, \sqrt{2})$$

where K are the terms independent of $m, \nabla m$. If $\lambda > B + \frac{L^2}{4}$ then we can find suitable ε, δ such that $\|m\|_{H_0^1(\Omega)}$ is bounded.⁷ After taking the difference of both sides, the inequality simplifies to

$$\tilde{\alpha} \|m\|_{H_0^1(\Omega)} \leq K$$

where $\tilde{\alpha}$ is the minimum of both factors of m^2 and ∇m^2 . □

⁷See Theorem 2.13 for further elaboration on this point.

Remark 3.3. This result has a very important implication. It tells us that any solution m is still bounded by a constant despite $G(\cdot)$ being non-linear.

Proposition 3.4. *If $\lambda > 0$ is sufficiently large, then the fixed point m is unique.*

Proof. Suppose m_1, m_2 both solve HJB. Taking the difference and setting $v = m_1 - m_2$ gives

$$\begin{aligned} \|\nabla(m_1 - m_2)\|_{L^2}^2 + \lambda \|m_1 - m_2\|_{L^2}^2 &\leq \int_{\Omega} \left| \frac{\partial H}{\partial p} \right| |m_1 - m_2| |\nabla(m_1 - m_2)| + |G_1 - G_2| |m_1 - m_2| dx \\ &\leq L \|m_1 - m_2\|_{L^2} \|\nabla(m_1 - m_2)\|_{L^2} + \tilde{B} \|m_1 - m_2\|_{L^2}^2 \end{aligned}$$

using Assumptions 3, 7 and Cauchy–Schwarz. Hence, by our previous use of Young’s inequality in Proposition 2.7,

$$\lambda - \tilde{B} > \frac{L^2}{4} \implies m_1 = m_2 \text{ in } H_0^1(\Omega).$$

□

As with the linear version of KFP we define a map $\tilde{S}$ that will now take the place of S in definition 2.17.

Definition 3.5. Define $\tilde{S} : H_0^1(\Omega) \rightarrow H_0^1(\Omega)$ by $\tilde{S}(u) = m$, where m solves non-linear KFP for given u .

Since the form of HJB in equation (3.1) is the same as that in equation (1.1), we can investigate the existence of a fixed point of the map $\tilde{A}$.

Definition 3.6. Let T and $\tilde{S}$ be as in definitions 2.17 and 2.18. Define

$$\tilde{A} : H_0^1(\Omega) \rightarrow H_0^1(\Omega), \quad \tilde{A} := \tilde{S} \circ T.$$

A fixed point of $\tilde{A}$ means a solution to the new system of equations.

Main Result for Non-linear KFP

Theorem 3.7 (Existence of MFG Solution with Non-linear KFP). *Let the Mean Field Game system be characterised by equation (3.1) and suppose the assumptions above are true. Then a solution $(u, m) \in H_0^1(\Omega) \times H_0^1(\Omega)$ exists.*

Proof. We prove the theorem via showing $\tilde{A}$ has a fixed point. Once again, we wish to use Schaefer’s theorem. We need to show that $\tilde{S}$ is continuous. Let $(u_k)_{k=1}^{\infty} \rightarrow u$ in $H_0^1(\Omega)$. Denote $m = \tilde{S}[u]$, $m_k = \tilde{S}[u_k]$, $b = \frac{\partial H}{\partial p}(x, \nabla u)$, $b_k = \frac{\partial H}{\partial p}(x, \nabla u_k)$ and $G_k = G(x, m_k)$. Taking the weak formulation of the difference in KFP and setting $v = m - m_k$ yields

$$\|\nabla(m - m_k)\|_{L^2}^2 + \lambda \|m - m_k\|_{L^2}^2 \leq \int_{\Omega} |\nabla(m - m_k)| |m b - m_k b_k| + |G - G_k| |m - m_k| dx.$$

Applying the Lipschitz condition (Assumption 7) to G, G_k gives

$$\|\nabla(m - m_k)\|_{L^2}^2 + \lambda \|m - m_k\|_{L^2}^2 \leq \int_{\Omega} |\nabla(m - m_k)| |m b - m_k b_k| + \tilde{B} |m - m_k|^2 dx.$$

By λ being sufficiently large to account for the $\tilde{B}$ term (Assumption 8) and recalling the same line of reasoning in Proposition 2.22 we conclude that $\tilde{S}$ is continuous. We also have that $\tilde{S}$ is bounded (Theorem 3.2) and that T is both compact (Proposition 2.29) and continuous (Proposition 2.21). $\tilde{A}$ is therefore compact.

To meet the final condition of Schaefer's theorem, we must show

$$\{m \in H_0^1(\Omega) : m = \mu \tilde{A}[m] \text{ for some } \mu \in [0, 1]\}$$

is bounded. The corresponding system, with $\tilde{A}[m]$ substituted for m/μ , is

$$-\Delta u + H(x, \nabla u) + \lambda u = f(x, m), \quad \frac{1}{\mu} \left(-\Delta m - \operatorname{div} \left(m \frac{\partial H}{\partial p}(x, \nabla u) \right) + \lambda m \right) = G(x, \frac{m}{\mu}).$$

Taking the weak form of the latter and setting $v = m$ implies

$$\begin{aligned} \int_{\Omega} |\nabla m|^2 + \lambda |m|^2 dx &\leq \int_{\Omega} \mu |G(x, \frac{m}{\mu})| |m| + L |m| |\nabla m| dx && \text{See Theorem 3.2} \\ &\leq \int_{\Omega} B(|m| + 1) |m| + L |m| |\nabla m| dx && (\text{Assumption 7}) \end{aligned}$$

In Theorem 3.2 we showed that when $\lambda > 0$ is sufficiently large, the inequality simplifies to

$$\tilde{\alpha} \|m\|_{H_0^1(\Omega)} \leq K$$

Thus, the set

$$\{m \in H_0^1(\Omega) : m = \mu \tilde{A}[m] \text{ for some } \mu \in [0, 1]\}$$

is bounded independently of μ . By Schaefer's theorem (2.12), $\tilde{A}$ has a fixed point. $\square$

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